

CYBERNAUT ED TECH PROJECT REPORT

HOSPITAL MANAGEMENT SYSTEM USING DJANGO

Submitted By

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PROJECT TITLE

HOSPITAL MANAGEMENT SYSTEM USING DJANGO

PROJECT DOMAIN

Web Application Development

Technology Stack

Python | Django | HTML | CSS | JavaScript

Submitted To

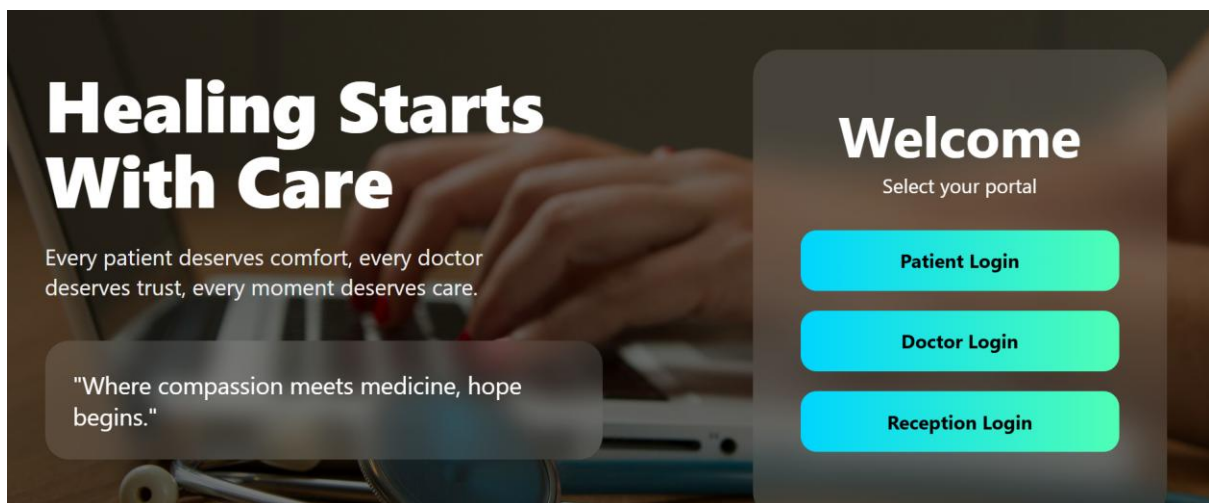
CYBERNAUT EDTECH

INTRODUCTION

The Hospital Management System is a comprehensive web-based healthcare management application developed using Python and Django Framework with the objective of digitizing and simplifying hospital operations. The system provides a centralized platform for managing appointments, prescriptions, billing, and user management while improving communication between patients, doctors, receptionists, and administrators.

In many hospitals and clinics, traditional methods still depend on manual paperwork and physical record maintenance. Appointment scheduling, patient registration, prescription management, and billing operations are often handled separately, which may lead to data duplication, delayed communication, increased waiting time, and difficulties in maintaining records efficiently. As the number of patients increases, managing hospital activities manually becomes more complex and time-consuming.

To overcome these challenges, the Hospital Management System was developed as a digital solution that integrates multiple hospital departments into a single platform. The application allows different users to access dedicated functionalities according to their roles through secure authentication and role-based access control.



The system consists of four major user modules:

Patient Module:

Patients are allowed to create their own accounts and access hospital services digitally. Patients can register, log in, book appointments with doctors, track appointment history, view uploaded prescriptions, and access electronic billing records.

Doctor Module:

Doctors are provided with a dedicated dashboard where they can view patient appointment requests, approve or reject appointments, manage consultation details, and upload prescriptions after completing patient consultation.

Receptionist Module:

The receptionist manages operational activities including monitoring appointments from all doctors and generating electronic bills for completed consultations. Billing information includes consultation fees, medicine charges, taxes, payment status, and transaction details.

Administrator Module:

The administrator controls the overall application and manages hospital users including doctors and receptionists. Administrative functions include user creation, monitoring activities, and maintaining system organization.

The Hospital Management System was developed using Django because of its powerful architecture, built-in authentication system, secure database management, and efficient handling of web applications. HTML and CSS were used to design modern and responsive user interfaces, while SQLite was used as the database for storing user information, appointments, prescriptions, and billing records.

The application follows a role-based workflow where patients initiate appointments, doctors process medical consultation and prescriptions, receptionists handle billing operations, and administrators supervise the overall system. This structured flow improves operational efficiency and ensures smooth coordination among departments.

This project also provided practical exposure to real-world software development concepts including frontend design, backend development, database modeling, authentication mechanisms, CRUD operations, file handling, dashboard development, debugging, version control using Git and GitHub, and project deployment practices.

Overall, the Hospital Management System demonstrates how modern web technologies can be applied in healthcare environments to improve service quality, reduce manual effort, increase data accessibility, and create a more efficient patient management experience.

2. OBJECTIVES

The objectives of this project are:

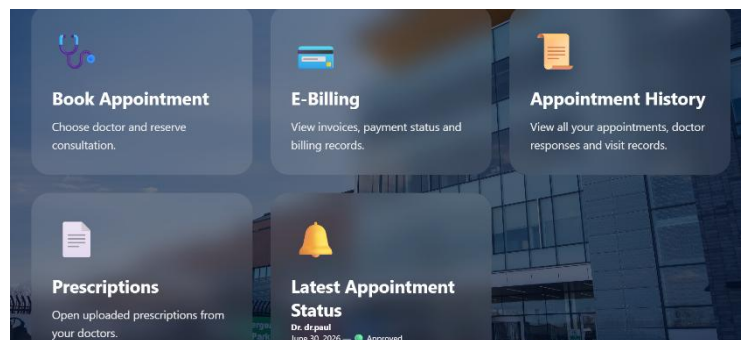
- Develop a complete hospital management web application.
- Implement secure authentication and role-based login.
- Allow patients to register and book appointments.
- Enable doctors to approve appointments and upload prescriptions.
- Allow receptionists to generate and manage bills.
- Provide administrators full control of hospital operations.
- Improve record management and reduce paperwork.
- Create an efficient healthcare workflow.

SYSTEM MODULES

A. PATIENT MODULE

Features:

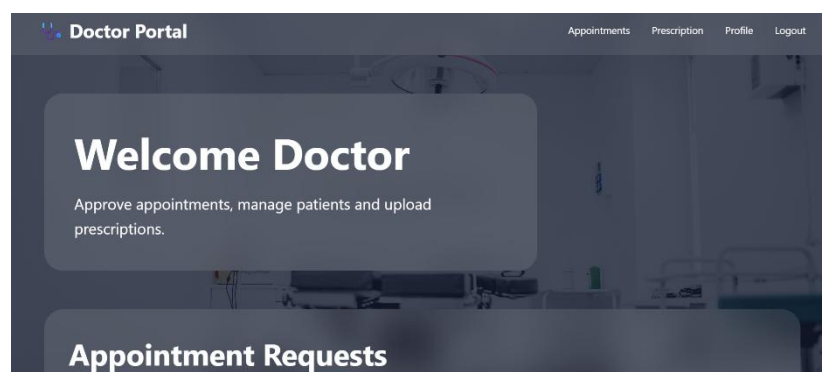
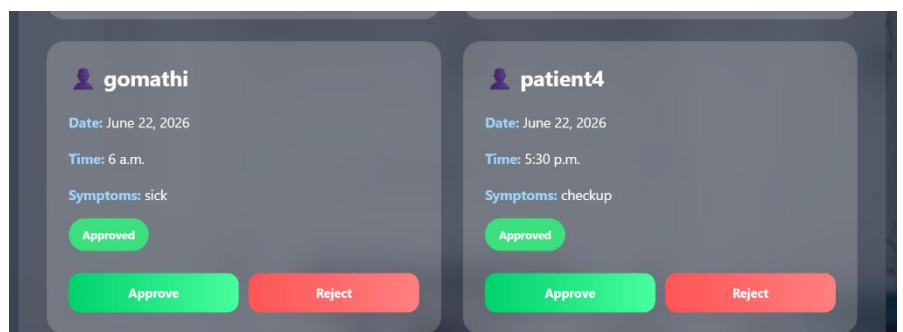
- ✓ Patient Registration
- ✓ Patient Login
- ✓ Book Appointment
- ✓ Appointment History
- ✓ View Prescriptions
- ✓ E-Billing



B. DOCTOR MODULE

Features:

- ✓ Doctor Login
- ✓ View Appointment Requests
- ✓ Approve Appointments
- ✓ Reject Appointments
- ✓ Upload Prescriptions
- ✓ Manage Patients



C. RECEPTIONIST MODULE

Features:

- ✓ Reception Login
- ✓ View Appointments
- ✓ Generate Bills
- ✓ Update Patient Records

The screenshot shows the 'MediCare Reception' interface. At the top, there's a blue header with the logo and 'Create E-Bill' and 'Logout' buttons. Below the header, the title 'Appointment Management' is displayed with a subtitle 'View appointments from all doctors and manage patient billing.' A table lists appointments with columns for Patient, Doctor, Date, Time, Symptoms, and Status. The table contains six rows of data.

Patient	Doctor	Date	Time	Symptoms	Status
patient4	Dr. di.paul	June 30, 2026	7 p.m.	fever	Approved
gomathi	Dr. di.paul	June 30, 2026	5:30 p.m.	fever	Approved
patient4	Dr. dr.smith	June 22, 2026	5:30 p.m.	checkup	Approved
patient4	Dr. di.paul	June 22, 2026	5:30 p.m.	sick	Pending
patient4	Dr. di.smith	June 22, 2026	5:30 p.m.	sick	Approved

D. ADMIN MODULE

Features:

- ✓ Admin Login
- ✓ Create Doctor Accounts
- ✓ Create Reception Accounts
- ✓ Manage Users
- ✓ Monitor System Activities

Generate E-Bill

Create payment invoice after patient payment.

Select Patient

Choose Patient

Consultation Fee

Medicine Fee

Tax

GENERATE BILL

SYSTEM WORKFLOW

Step 1:

Patient creates account



Step 2:

Patient logs in



Step 3:

Patient books appointment



Step 4:

Doctor receives request



Step 5:

Doctor approves/rejects



Step 6:

Doctor uploads prescription



Step 7:

Receptionist generates bill



Step 8:

Patient views prescription and billing

FEATURES IMPLEMENTED

Authentication

- ✓ Patient Login
- ✓ Doctor Login
- ✓ Reception Login
- ✓ Admin Login

Patient Features

- ✓ Appointment Booking
- ✓ History
- ✓ Prescription Access

Doctor Features

- ✓ Approval System
- ✓ Prescription Upload

Reception Features

- ✓ Billing Management

Admin Features

- ✓ User Management

ADVANTAGES

- Easy to use
- Reduces paperwork
- Faster appointment scheduling
- Centralized database
- Better record management
- Secure access
- Role-based control

FUTURE ENHANCEMENTS

- Online Payment
- SMS Notifications
- Email Alerts
- AI Chatbot
- Telemedicine
- Cloud Deployment
- Analytics Dashboard

CONCLUSION

The Hospital Management System developed using Django successfully digitizes hospital workflows and improves patient management, appointment scheduling, prescription handling, and billing operations.

The application demonstrates practical implementation of full-stack web development concepts and provides an efficient and